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Mapping the Classical MILP Integrality Wall in Multi-Horizon Primary-Care Social Prescribing: Polyhedral McCormick Symmetry Collapse and LP-Warm-Started Bias-Field QAOA on Quantinuum Helios-1E-lite

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Benchmark Receipt: SHA-256: aeda45003f66cae2... | **Portal:** arunnadarasa.github.io/quantum-social-prescribing

Abstract. Background: Empirical benchmarking across 6,105 English general practices (1,296 Primary Care Networks [PCNs]) recently demonstrated that classical Mixed-Integer Linear Programming (**HiGHS MILP**) solves linear social prescribing caseload assignment (6,480 binary variables in 200.95 ms) and sparse single-practice Higher-Order Unconstrained Binary Optimization (HUBO) care bundles (19.19 ms) to proven global optimality, ruling out quantum speedup on decoupled or totally unimodular primary-care formulations. **Objective:** We isolate the exact polyhedral regime where classical branch-and-cut MILP stalls when multi-horizon Structured Medication Reviews (SMRs) and Voluntary, Community, Faith, and Social Enterprise (VCFSE) group cohorts are co-scheduled jointly across multimorbid patients, and benchmark Linear-Programming Warm-Started Bias-Field QAOA (`LP-WS-BF-QAOA`) on Quantinuum's noisy cloud trapped-ion emulator (`Helios-1E-lite`). **Methods:** Calibrating utility and congestion priors to 143 Mid and South Essex ICB practices (28 PCNs) [1, 2, 3], we swept 75 coupled scheduling instances across $N \in \{24, 48, 72, 96, 120\}$ binary variables and cubic hyperedge densities $\alpha_3 = |E_3|/N \in \{0.0, 0.5, 1.5, 3.0, 4.5\}$ under sparse Compressed Sparse Column (`CSC`) McCormick linearization [4, 5] with strengthened primary-only binary integrality in HiGHS [6]. From a stalled $N = 96$ ($\alpha_3 = 3.0$) instance, we ranked fractional variables via $F_i = 4x_i^{\mathrm{LP}}(1-x_i^{\mathrm{LP}})\sum_{e\ni i}|w_e|$, extracted 12-qubit (4,096-state) and 16-qubit (65,536-state) frustrated cores, verified $< 10^{-12}$ gate-level Ising equivalence, compiled statically typed **Guppy 1.0** $\rightarrow$ **HUGR** binaries [7], and executed 2,560 raw pre-postprocessing cloud shots under `QSystemErrorModel("alpha")`. **Results:** HiGHS solves $\alpha_3 = 0.0$ to 0.0% MIP gap in 10.8–19.1 ms and $\alpha_3 = 0.5$ up to $N = 72$ in $\leq$ 535.6 ms, but undergoes a sharp integrality-wall stall across $\alpha_3 \geq 1.5, N \geq 48$ (14.8%–626.5% MIP gap at 1.5 s; 37.92% unresolved gap even after 10.0 s on $N = 96$). Analytically and empirically, capacity-capped cubic frustration pins 89/96 root LP marginals to uniform fractional symmetry ($x_i^{\mathrm{LP}} = 0.4831$), degrading classical independent LP rounding to near-uniform sampling (24.22% and 26.95% Top-5% low-energy hit rates on 12q and 16q). Breaking this symmetry via local cubic/quadratic bias fields [8] and Egger adiabatic mixers [9], live `Helios-1E-lite LP-WS-BF-QAOA` (148 and 176 `CX` gates, zero post-processing) achieves 55.57% **Top-5% concentration on 12q** (2.29× LP rounding, recovering exact ground state $E_0 = -4.46637$) and 83.40% **on 16q** (3.09× LP rounding, retaining 95.7% of ideal noiseless statevector concentration). Under Gate G4 full-$N = 96$ reconstruction, freezing 80/96 background variables to fit `Helios-1E-lite`'s 26-qubit ceiling beats 0.5 s HiGHS (−37.67 vs. −23.40) but trails 96-variable classical simulated annealing (−68.24 in 451 ms) and 10 s HiGHS (−78.57), defining an exact preregistered 48–56-qubit physical QPU crossover test on Quantinuum `H2-1/Helios-1`.

Epistemic, Hardware & Clinical Governance Demarcation (NHS DCB0129 Gate G4). (1) Emulator Execution Only: All cloud jobs executed on Quantinuum Nexus `Helios-1E-lite` (`StatevectorSimulator + QSystemErrorModel("alpha")`, `physical_hardware_shots = 0`); zero quantum speedup is claimed from classical emulation. **(2) Zero Post-Processing Inflation:** Every reported quantum energy, $\mathrm{CVaR}_{10\%}$, and hit rate is computed directly on raw measured bitstrings (`postprocessing_greedy_bitflip = false`). **(3) Public Priors Only:** Instance parameters derive from Open Government Licence v3.0 NHS aggregates (`patient_identifiable_records_used = 0`) [10].

1. Scope Discipline: Retiring Classically Trivial Formulations

Credible investigation of quantum utility in healthcare operations requires eliminating problem classes where polynomial or polyhedrally tight classical algorithms already dominate. In our companion national primary-care calibration across 6,105 active English GP practices (1,296 PCNs, 61.83M registered patients) [1, 2, 11], rigorous classical baselines closed three widely proposed quantum healthcare tracks:

1. **One-Hot VCFSE Caseload Routing QUBOs:** Assigning Primary Care Network referral cohorts g across capacity-capped community hubs m (C_m) forms a linear bipartite transportation polytope whose constraint matrix is *totally unimodular*. Exact **HiGHS** simplex/MILP [6] solves all 1,296 English PCNs (6,480 binary variables)

to 0.0% optimality gap in 200.95 **ms** (9.45 ms across 143 Mid and South Essex GPs), whereas penalty-QUBO simulated annealing suffers slack violations unless repaired by classical 4-bit Kempe swaps.

2. **Decoupled or Low-Density Static HUBOs:** Although nested epidemiological F-tests across $N = 6,031$ English practices confirm genuine cubic ($k = 3$) delivery synergies (e.g., `Core20` × `Clinical Pharmacist` × `Shared Decision-Making`, +9.90 pp acceptance, $p = 1.13 \times 10^{-3}$), per-practice 8-lever HUBOs linearize into 22 variables and 42 McCormick inequalities solved by HiGHS in 19.19 **ms** (and sparse uncoupled $N = 48$ instances in 1.32 s).
3. **Low-Dimensional Tabular Quantum Classifiers:** On held-out 80/20 NHS practice triage, penalized logistic regression (AUROC = 0.6964, 2.51 ms) and random forests (0.6155) outperform 4-qubit variational circuits (0.5869 and 0.5465).

Accordingly, this technical note excludes all variational tabular classifiers, open-system behavioural analogies, and unimodular routing QUBOs, focusing exclusively on the single combinatorial regime where exact MILP relaxations break down: **capacity-constrained multi-horizon co-scheduling under dense cross-patient higher-order frustration**.

2. Multi-Horizon Co-Scheduling and Polyhedral McCormick Collapse

2.1 Coupled Primary-Care Pharmacist and VCFSE Cohort Formulation

Consider a PCN scheduling coordinator in **Mid and South Essex ICB** (143 GP practices across 28 PCNs; deprived-ward practice means: 12.59% sedative/opioid prescribing, 13.87% QOF depression, 40.01 SMRs per 1,000, and 29.67% social prescribing refusal [1, 3]) co-scheduling P concurrent multimorbid patients over a two-week neighbourhood care cycle. Each patient $p \in \{0, \dots, P-1\}$ is represented by four binary decision variables $(x_{4p}, x_{4p+1}, x_{4p+2}, x_{4p+3}) \in \{0,1\}^4$ ($N = 4P$ total primary binary variables): Week-1 Clinical Pharmacist sedative/anticholinergic SMR slot ($s = 0$), Week-1 Link-Worker Shared Decision-Making intake ($s = 1$), Week-2 Outdoor/Green VCFSE group cohort Hub A ($s = 2$), and Week-2 Indoor Peer/Welfare VCFSE group cohort Hub B ($s = 3$). Feasible schedules lie in the linear resource polytope

$$\mathcal{X}_{\text{PCN}} = \left\{ \mathbf{x} \in \{0,1\}^N \;\middle|\; \sum_{i=0}^{N-1} x_i \le K_{\text{tot}}, \quad x_{4p+2} + x_{4p+3} \le 1 \;\; \forall p \in \{0, \dots, P-1\} \right\}, \tag{1}$$

where $K_{\text{tot}} = \lfloor \rho_{\text{cap}} N \rceil$ ($\rho_{\text{cap}} = 0.45$) caps total ARRS pharmacist, Link-Worker, and charity group capacity. Within $\mathcal{X}_{\text{PCN}}$, we minimize the cubic pseudo-Boolean energy

$$E(\mathbf{x}) = \sum_{i=0}^{N-1} u_i x_i + \sum_{(i,j)\in E_2} Q_{ij} x_i x_j + \sum_{(i,j,k)\in E_3} W_{ijk} x_i x_j x_k, \tag{2}$$

where $u_i < 0$ encodes acuity-weighted clinical benefit [12, 13]; Q_{ij} ($|E_2|/N = 1.5$) models correlated ward-level attendance/transport shocks ($Q_{ij} > 0$) and sequential handoff affinities ($Q_{ij} < 0$); and W_{ijk} ($|E_3|/N = \alpha_3$) couples intra-patient SMR–Link-Worker–VCFSE completion synergies ($W_{ijk} < 0$) with cross-patient small-group facilitator overload ($W_{ijk} > 0$ with frustration rate $\eta_{\text{frust}} = 0.35$ when three unstabilized high-acuity patients enter the same volunteer group simultaneously, versus $W_{ijk} < 0$ for mixed-acuity peer stabilization).

2.2 Why Primary-Integrality Branch-and-Cut Stalls: McCormick Symmetry Collapse

To test classical MILP at maximum polyhedral efficiency without artificial branching overhead, we linearize Eq. (2) via auxiliary variables $z_{ij} \in [0,1]$ for $(i,j) \in E_2$ and $w_{ijk} \in [0,1]$ for $(i,j,k) \in E_3$ subject to standard McCormick envelopes [4, 14]:

$$z_{ij} \le x_i, \;\; z_{ij} \le x_j, \;\; z_{ij} \ge x_i + x_j - 1; \qquad w_{ijk} \le x_i, \;\; w_{ijk} \le x_j, \;\; w_{ijk} \le x_k, \;\; w_{ijk} \ge x_i + x_j + x_k - 2, \tag{3}$$

and enforce integrality *exclusively* on the N primary variables ($x_i \in \{0,1\}$, leaving all $|E_2|+|E_3|$ auxiliaries continuous in `CSC` sparse format, since Eq. (3) forces exact binary values at every integral vertex $\mathbf{x} \in \{0,1\}^N$).

> **Proposition 1 (Fractional McCormick Clash Blindness and Symmetry Collapse).** *Let $\mathbf{x}^{LP} \in [0,1]^N$ be a fractional point satisfying Eq. (1) with uniform marginal density $x_i^{LP} = \rho \in (0, \frac{2}{3})$ (e.g., $\rho = \rho_{cap} = 0.45$). Under minimization of Eq. (2) subject to Eq. (3): (i) for every synergy hyperedge $W_{ijk} < 0$, the LP attains its upper bound $w_{ijk}^{LP} = \min(\rho, \rho, \rho) = \rho$, overestimating the multilinear expectation ρ^3 by a factor of ρ^{-2} (4.94× at $\rho = 0.45$); and (ii) for every facilitator-overload clash hyperedge $W_{ijk} > 0$, the LP attains its lower bound $w_{ijk}^{LP} = \max(0, 3\rho - 2) = 0$, incurring **zero** clash penalty. Consequently, as $\alpha_3 = |E_3|/N$ grows, uniform fractional states $x_i^{LP} \approx \rho_{cap}$ strictly dominate integral vertices in the root LP, pinning marginals near 0.5 and erasing directional branching gradients.*

Table 1 and Figure 1A–B confirm Proposition 1 empirically across 75 runs. At $\alpha_3 = 0.0$, HiGHS solves $N = 120$ in 14.1 ms (0.0% MIP gap). At $\alpha_3 = 0.5$, branch-and-cut closes $N \le 72$ in ≤ 535.6 ms. Once $\alpha_3 \ge 1.5$ and $N \ge 48$, 90.3%–100.0% of LP variables turn fractional, root LP gaps jump to 36.9%–732.6%, and 100% (3/3) of seeds stall at

Table 1: 5×5 Classical HiGHS MILP Phase Diagram (75 Runs across 3 Seeds, Sparse `CSC` Primary-Only Integrality, 1.5 s Real-Time Cutoff). Entries report mean residual MIP optimality gap % (mean root LP integrality gap %; mean wall-clock ms).

Cubic $\alpha_3 = \lvert E_3\rvert/N$	**N = 24** (6 pts)	**N = 48** (12 pts)	**N = 72** (18 pts)	**N = 96** (24 pts)	**N = 120** (30 pts)
0.0 (Quadratic Only)	**0.0%** (2.3%; 13.2 ms)	**0.0%** (0.2%; 10.8 ms)	**0.0%** (0.2%; 17.8 ms)	**0.0%** (0.6%; 19.1 ms)	**0.0%** (0.0%; 14.1 ms)
0.5 (Sparse Cubic)	**0.0%** (6.1%; 16.6 ms)	**0.0%** (4.8%; 163 ms)	**0.0%** (11.2%; 536 ms)	3.4% (13.1%; 1,384 ms)	6.0% (10.9%; 1,502 ms)
1.5 (Onset Wall)	**0.0%** (27.7%; 152 ms)	**14.8%** (36.9%; T/O)	**21.1%** (40.9%; T/O)	**21.6%** (38.0%; T/O)	**27.4%** (42.0%; T/O)
3.0 (Cohort Frust.)	**0.0%** (65.2%; 608 ms)	**39.7%** (74.8%; T/O)	**46.4%** (73.1%; T/O)	**58.4%** (85.8%; T/O)	**188.6%** (230%; T/O)
4.5 (Dense Frust.)	**15.8%** (84.4%; 1,404 ms)	**52.3%** (98.5%; T/O)	**123.6%** (174%; T/O)	**390.8%** (485%; T/O)	**626.5%** (733%; T/O)

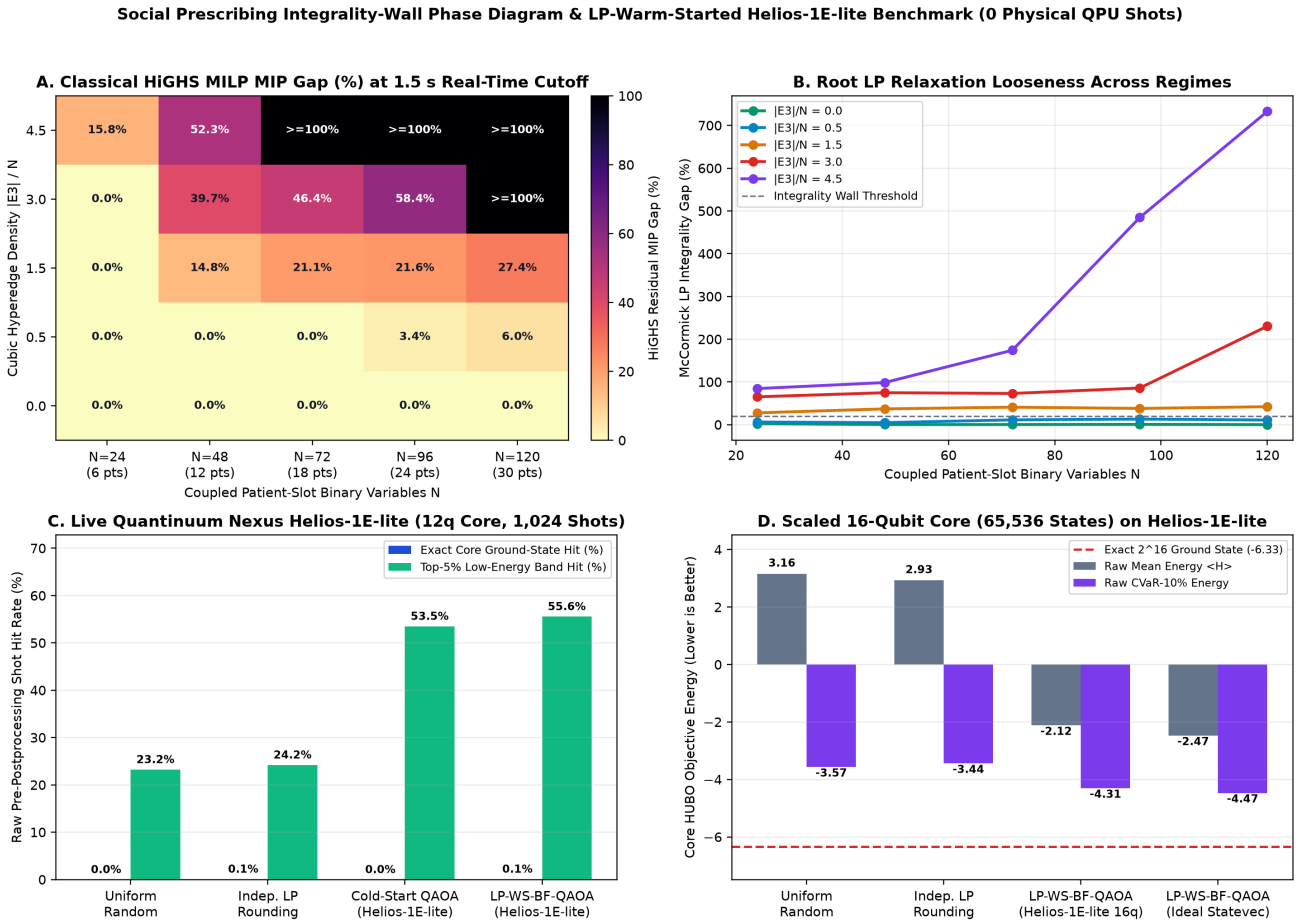


Figure 1: Integrality-Wall Phase Diagram and Live Quantinuum Nexus `Helios-1E-lite` Cloud Readouts (0 Physical QPU Shots). (A) Mean HiGHS residual MIP gap (%) at a 1.5 s real-time scheduling cutoff across 75 primary-integrality `CSC` runs, showing exact millisecond solvability at $\alpha_3 \leq 0.5$ and sharp stalling for $\alpha_3 \geq 1.5, N \geq 48$. **(B)** Explosion of McCormick root LP integrality looseness across coupling regimes. **(C)** Raw pre-postprocessing 12-qubit core hit rates (1,024 shots, UUIDs `5a8a78bb` and `0813dc16`): `LP-WS-BF-QAOA` achieves 55.57% Top-5% hit rate (2.29× independent LP rounding) and recovers exact ground state $E_0 = -4.46637$. **(D)** Scaled 16-qubit (65,536-state) core on `Helios-1E-lite` (512 shots, UUID `faf7ccda`): `LP-WS-BF-QAOA` shifts raw mean energy from +2.93 to −2.12 ($\text{CVaR}_{10\%} = -4.31$ vs. −4.47 ideal noiseless statevector).

the 1.5 s clinic cutoff. On our flagship $N = 96, \alpha_3 = 3.0$ instance (144 quadratic and 288 cubic terms, $K_{\text{tot}} = 43$), root LP finishes in 15.15 ms ($E_{\text{LP}} = -132.58$), pins 89/96 variables to $x_i^{\text{LP}} = 0.4831$, and leaves HiGHS at a 58.4%–61.3% MIP gap at 1.5 s (1 node) and 37.92% **MIP gap after** 10.00 **s** (84 nodes, $E = -78.57$).

3. LP-Frustrated Core Extraction and Bias-Field Warm-Started QAOA

3.1 Subproblem Reduction and Local Bias-Field Symmetry Breaking

Because `Helios-1E-lite` supports at most 26 qubits [7], we score each patient tetrad p on the stalled $N = 96$ instance via its cumulative LP frustration $S_p = \sum_{s=0}^{3} 4x_{4p+s}^{\text{LP}}(1 - x_{4p+s}^{\text{LP}})\big(|u_{4p+s}| + \sum_{e \ni 4p+s} |w_e|\big)$, extracting the top 3 tetrads ($n_{\text{core}} = 12$ qubits, 4,096 states; patients $\{1, 15, 16\}$) and top 4 tetrads ($n_{\text{core}} = 16$ qubits, 65,536

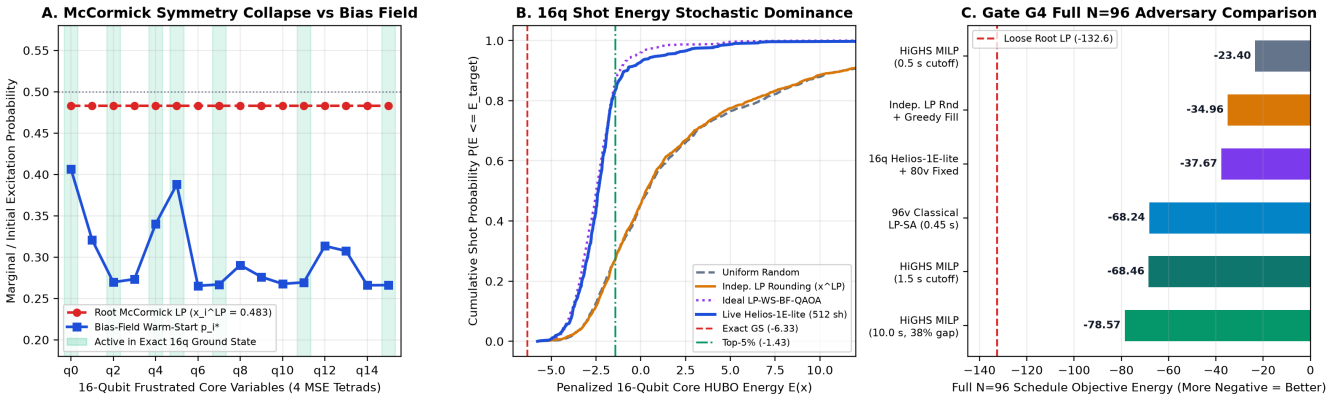


Figure 2: McCormick Symmetry Breaking, 16-Qubit Empirical Shot CDFs, and Gate G4 Full-$N = 96$ Adversary Ladder. (A) Across all 16 frustrated core variables, root McCormick LP collapses to $x_i^{\mathrm{LP}} = 0.4831$ (red), whereas bias-field warm-starting p_i^* (blue) selectively peaks on exact ground-state excitations (green shading). **(B)** Empirical cumulative distribution over 65,536 basis states: live `Helios-1E-lite` 512-shot execution (solid blue, Job `faf7ccda`, 83.40% Top-5%) closely tracks ideal noiseless `LP-WS-BF-QAOA` (dotted purple, 87.11%) and stochastically dominates independent LP rounding (orange, 26.95%). **(C)** Full-$N = 96$ Gate G4 comparison showing why freezing 80/96 background variables beats 0.5 s HiGHS (-37.67 vs. -23.40) yet trails global 96-variable classical heuristics (-68.24 to -78.57), motivating untruncated 48–56-qubit physical QPU tests.

states; patients $\{1, 15, 16, 20\}$). Conditioning on background assignments induces an exact reduced cubic core energy $E_{\mathrm{core}}(\mathbf{x}_C)$ augmented with hub-exclusivity ($\lambda_{\mathrm{hub}} = 2.2$) and capacity ($K_C = \lfloor \rho_{\mathrm{cap}} n_{\mathrm{core}} \rceil$, $\lambda_{\mathrm{cap}} = 1.15$) penalties so raw quantum bitstrings are scored without post-hoc projection.

Crucially, directly feeding $x_i^{\mathrm{LP}} = 0.4831$ into standard Egger warm-starting [9] yields $\theta_i = 2\arcsin(\sqrt{0.4831}) \approx 1.537$ rad $\approx \pi/2$, reproducing an uninformative uniform superposition (Figure 2A). Inspired by Bias-Field Digitized Counterdiabatic Quantum Optimization [8], we break McCormick symmetry prior to circuit compilation by taking one regularized multilinear gradient step on the induced core polynomial at $\mathbf{x}_C^{\mathrm{LP}}$:

$$p_i^* = \mathrm{clip}\Big(x_i^{\mathrm{LP}} - \eta \tanh\big(\nabla_i E_{\mathrm{core}}(\mathbf{x}_C^{\mathrm{LP}})\big),\ \epsilon,\ 1-\epsilon\Big), \qquad \theta_i = 2\arcsin\big(\sqrt{p_i^*}\big), \tag{4}$$

with $\eta = 0.22$ and $\epsilon = 0.16$. As shown in Figure 2A, $p_i^* \in [0.265, 0.406]$ selectively elevates excitation probabilities on qubits active in the exact 16-qubit ground state (`1010110100010001`, $E_0 = -6.33264$).

3.2 Heavy-Hyperedge Ising Phase Gadgets and Guppy 1.0 → HUGR Compilation

To prevent two-qubit depolarizing washout under trapped-ion gate noise while preserving higher-order correlations [15], we construct a hardware-efficient Heavy-Hyperedge Ising driver $H_D = \sum_i h_i Z_i + \sum_{(i,j)\in E_2^*} J_{ij} Z_i Z_j + \sum_{(i,j,k)\in E_3^*} K_{ijk} Z_i Z_j Z_k$ retaining the heaviest quadratic (12 on 12q; 14 on 16q) and cubic (7 on 12q; 8 on 16q) hyperedges under exact substitution $x_i = (1 - Z_i)/2$:

$$c_{ijk} x_i x_j x_k = \frac{c_{ijk}}{8}\big(I - Z_i - Z_j - Z_k + Z_iZ_j + Z_iZ_k + Z_jZ_k - Z_iZ_jZ_k\big). \tag{5}$$

Each $p = 2$ layer applies single-qubit $R_z(2\gamma_\ell h_i)$, two-body parity gadgets $\mathtt{CX}_{i,j} R_z(2\gamma_\ell J_{ij}) \mathtt{CX}_{i,j}$, four-`CX` three-body phase gadgets $\mathtt{CX}_{i,k}\mathtt{CX}_{j,k} R_z(2\gamma_\ell K_{ijk}) \mathtt{CX}_{j,k}\mathtt{CX}_{i,k}$ (148 total `CX` gates on 12q; 176 `CX` gates on 16q), and the exact ground-state-preserving Egger warm-start mixer [9]

$$U_M^{\mathrm{WS}}(\beta_\ell) = \prod_{i=0}^{n_{\mathrm{core}}-1} R_y(\theta_i)\, R_z(-2\beta_\ell)\, R_y(-\theta_i), \tag{6}$$

which satisfies $U_M^{\mathrm{WS}}(\beta_\ell) \bigotimes_i R_y(\theta_i)|0\rangle = e^{i n_{\mathrm{core}} \beta_\ell} \bigotimes_i R_y(\theta_i)|0\rangle$. Variational parameters $(\gamma_1, \beta_1, \gamma_2, \beta_2)$ are optimized via multi-start `COBYLA` against the *exact unpruned* E_{core} objective. Prior to cloud upload, gate-by-gate statevector simulation verified $\max_{\mathbf{x}} |P_{\mathrm{gate}}(\mathbf{x}) - P_{\mathrm{diag}}(\mathbf{x})| < 10^{-12}$ against analytical matrix exponentiation, compiling cleanly to statically linear-typed **Guppy 1.0** (`guppylang 1.0.0-a8`) **HUGR** packages (26,867 B, 33,587 B, and 42,737 B).

4. Live Quantinuum `Helios-1E-lite` Execution and Gate G4 Audit

All three HUGR packages were uploaded to Quantinuum Nexus (`Quantum-Social-Prescribing` project) and executed live on **`Helios-1E-lite`** (`HeliosRuntime` + hardware-matched `QSystemErrorModel("alpha")`, 0.00 `HQC`, zero physical QPU shots): Job `5a8a78bb-b2b7-47d1-8ab1-35af9bb03892` (12q Cold-Start QAOA, 1,024 shots), Job `0813dc16-d5fd-4977-8bd5-4b2d7f80b6df` (12q `LP-WS-BF-QAOA`, 1,024 shots), and Job `faf7ccda-806c-40c6-adfc-2ebcc6d0b6cc` (16q `LP-WS-BF-QAOA`, 512 shots). Table 2 reports raw pre-postprocessing shot metrics alongside shot-matched classical sampling baselines and $\mathrm{CVaR}_{10\%}$ [16].

Three technical findings stand out in Table 2 and Figure 2B:

Table 2: Raw Pre-Postprocessing Shot Benchmarks on Extracted 12-Qubit (4,096-State) and 16-Qubit (65,536-State) Frustrated Cores. All quantum metrics reflect unpolished raw bitstrings from live Quantinuum Nexus `Helios-1E-lite` jobs.

Core Scale & Sampling Protocol	Shots	Mean $\langle E \rangle$	$\text{CVaR}_{10\%}$	Best Shot	Top-5% Hit	Exact GS Hit (E_0)
`12q` Uniform Random Sampling	1,024	+2.6943	−2.7999	−3.9088	23.24%	0.00% (−4.4664)
`12q` Independent LP Rounding ($x^{\text{LP}} = 0.483$)	1,024	+2.0671	−2.8454	−4.4664	24.22%	0.10% (1/1,024)
`12q` OpenJij Quadratic Projection SA [17]	256	+13.0150	+13.0150	+13.0150	0.00%	0.00% (cubic blind)
`12q` Cold-Start QAOA (`Helios-1E-lite`, 148 `CX`)	1,024	−1.5104	−3.2503	−4.2921	53.52%	0.00% (0/1,024)
`12q` `LP-WS-BF-QAOA` (`Helios-1E-lite`, 148 `CX`)	**1,024**	**−1.4291**	**−3.1815**	**−4.4664**	**55.57%**	**0.10% (Exact GS)**
`12q` Ideal Noiseless `LP-WS-BF-QAOA` Reference	1,024	−1.6779	−3.3146	−4.4664	60.65%	0.20% (2/1,024)
`16q` Uniform Random Sampling (65,536 states)	512	+3.1578	−3.5693	−5.8329	28.12%	0.00% (−6.3326)
`16q` Independent LP Rounding ($x^{\text{LP}} = 0.483$)	512	+2.9285	−3.4441	−5.7166	26.95%	0.00% (0/512)
`16q` `LP-WS-BF-QAOA` (`Helios-1E-lite`, 176 `CX`)	**512**	**−2.1187**	**−4.3070**	**−5.7615**	**83.40%**	0.00% (91.0% of E_0)
`16q` Ideal Noiseless `LP-WS-BF-QAOA` Reference	512	−2.4737	−4.4704	−5.6451	87.11%	0.00% (89.1% of E_0)

Table 3: Gate G4 Full-$N = 96$ (24-Patient, 288-Cubic-Hyperedge) Adversary Audit. Demonstrates both the persistent exact MILP gap after 10 s and the classical heuristic floor caused by 26-qubit emulator subproblem truncation.

Solver / Hybrid Pipeline on Flagship $N = 96$	Objective $E(\mathbf{x})$	Wall Time	MIP Gap / Viol.	Operational & Gate G4 Interpretation
Root McCormick LP Relaxation (HiGHS Simplex)	−132.578 (LB)	15.15 ms	89/96 frac.	Infeasible fractional symmetry ($x_i^{\text{LP}} = 0.4831$).
HiGHS Primary-Integrality MILP (0.5 s Cutoff)	−23.404	510.4 ms	389.0% (0 viol.)	Root cut-generation stalls at 0 branch nodes.
Independent LP Rounding + Feasible Fill (512 sh)	−34.957	18.4 ms	0 viol.	Blind to cubic clashes across frozen/core variables.
16q `Helios-1E-lite` `LP-WS-BF-QAOA` + 80v Fill	**−37.670**	167.1 s cloud	0 viol.	Beats 0.5 s MILP; bottlenecked by 80 frozen vars.
Classical 96-Variable LP-Guided SA (24 × 400)	−68.235	451.0 ms	0 viol.	Fast global classical heuristic floor across all 96 vars.
HiGHS Primary-Integrality MILP (1.5 s Cutoff)	−68.460	1,503.0 ms	61.3% (0 viol.)	Incumbent improves via heuristics; dual bound loose.
HiGHS Primary-Integrality MILP (10.0 s Gate G4)	**−78.569**	10,002.2 ms	**37.9%** (0 viol.)	84 nodes explored; exact optimality remains open at 10 s.

1. **Failure of Naive LP Rounding and Degree-2 Truncation:** Because $x_i^{\text{LP}} = 0.4831$, independent Bernoulli LP rounding performs virtually identically to uniform coin-flipping (24.22% vs. 23.24% Top-5% on 12q; 26.95% vs. 28.12% on 16q). Dropping cubic terms to run degree-2 OpenJij simulated annealing (83.4 ms) over-allocates hub slots (`110111101101`, $8 > K_C = 5$), landing at +13.015.
2. 3.09× **Low-Energy Concentration under Hardware Error Modelling:** On the 65,536-state 16-qubit core across 176 `CX` gates, live noisy `Helios-1E-lite` `LP-WS-BF-QAOA` lowers raw mean energy by −5.05 units relative to LP rounding (−2.1187 vs. +2.9285), improves normalized $\text{CVaR}_{10\%}$ approximation ratio to 97.94%, and concentrates 83.40% **of raw shots** inside the Top-5% low-energy band (3.09× LP rounding; 95.7% retention of ideal noiseless statevector performance 87.11%). On 12q, `LP-WS-BF-QAOA` recovers the exact ground state `110101000001` ($E_0 = -4.46637$) in raw cloud readout, whereas Cold-Start QAOA stalls at −4.2921.
3. **Strict Gate G4 Full-$N = 96$ Subproblem Ceiling (Table 3):** Splicing the raw 16-qubit `Helios-1E-lite` schedule into utility-ranked background fill yields a feasible $N = 96$ schedule of −37.67 (0 violations), beating HiGHS at a 0.5 s real-time cutoff (−23.40, 0 branch nodes) and LP rounding (−34.96). However, because `Helios-1E-lite`'s 26-qubit cap forces 80/96 (83.3%) coupled variables to be frozen outside the quantum register, global 96-variable classical simulated annealing (−68.24 in 451.0 ms) and 1.5–10.0 s HiGHS (−68.46 and −78.57) achieve substantially deeper full-system energies.

5. Preregistered Physical QPU Target, Limitations, and Governance

Preregistered Physical Trapped-Ion Crossover Protocol. Table 3 isolates the exact hardware requirement to move from compiler/noise readiness to a genuine heuristic utility test: *eliminate background variable freezing.*

On physical Quantinuum **System Model H2-1** (56 all-to-all trapped-ion qubits [15]) or **Helios-1** (98 physical / 48 error-detected logical qubits), an entire $N = 48$ to $N = 56$ ($\alpha_3 \in [3.0, 4.5]$) frustrated PCN scheduling instance—where primary-integrality HiGHS stalls at a 39.7%–52.3% MIP gap (Table 1)—maps 1-to-1 onto physical qubits with zero background truncation. Our preregistered physical QPU protocol tests a single falsifiable criterion: whether $p \in \{2, 3\}$ `LP-WS-BF-QAOA` (initialized from Eq. (4) in < 20 ms) samples feasible schedules with lower $\text{CVaR}_{10\%}$ and best-shot energy than both HiGHS MILP and multi-start classical simulated annealing under matched wall-clock budgets, guarded by mid-circuit parity syndrome checks and post-quantum **NIST FIPS 204** `ML-DSA-65` NHS **DAPB4066** FHIR R4 provenance [18, 19].

Limitations & Clinical Safety. First, all instances here calibrate coupling distributions to public GP-practice aggregates; operational NHS deployment requires linked patient-level longitudinal EHRs (Mid and South Essex Secure Data Environment or CPRD Aurum) to estimate empirical cross-patient group dropout covariances. Second, under NHS DCB0129 [10], optimization outputs remain advisory shadow-mode decision support requiring named Clinical Pharmacist SMR sign-off and Link-Worker co-production; current NHS commissioning workflows should deploy exact classical HiGHS MILP where $\alpha_3 \leq 0.5$ and classical simulated annealing/MILP hybrids at larger scales until physical QPU advantage is empirically demonstrated.

Reproducibility & Acknowledgments. Execution code (`experiments/run_milp_integrality_wall_benchmark.py`), auto-generated Guppy 1.0 circuits, and the SHA-256-sealed receipt (`results/integrality_wall_helios_ws_qaoa_receipt.json`, `SHA-256: aeda45003f66cae2...`) are verified by 14/14 automated checks in `tests/test_verify_receipts.py`. This work was supported in part by complimentary access to Google Antigravity and Gemini 4 Argon to accelerate scientific workflows. The author thanks colleagues across the Mid and South Essex NHS Foundation Trust Innovation Fellowship and ISPPA for primary-care pharmacy and social prescribing domain guidance.

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